

Question 2

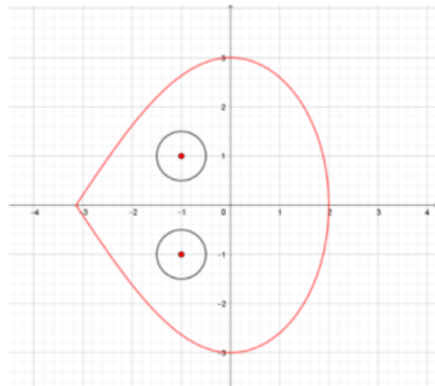
Complete
Mark 1.750 out of 3.000
Flag question

Na horta EcoCampus ISEC a D. Toupeira tem dito presente e apreciado o trabalho feito naquele espaço do campus do ISEC.

A figura seguinte representa a projeção da cabeça da toupeira no plano xOy, limitada pelas funções f e g e com uns olhos circulares de raio $1/2$.

$$f(x) = \begin{cases} 3\sqrt{1 - \frac{x^2}{4}} & , 0 < x \leq 2 \\ 3\cos(\frac{x}{2}) & , -\pi \leq x \leq 0 \end{cases}$$

$$g(x) = -f(x)$$



a) Defina polinómio interpolador.

b) Aplicando a interpoladora de Newton das diferenças divididas, determine os polinómios interpoladores de grau 2 das funções $f(x)$ para $x \in [-\pi, 0]$ e $g(x)$ para $x \in [0, 2]$.

c) Sem calcular, estabeleça em coordenadas cartesianas o integral duplo que lhe permitiria determinar a média de área da figura excluindo os círculos.

Resolvido em Papel

Question 3

Partially correct
Mark 0.600 out of 3.000
Flag question

Considere a equação diferencial $(yx^2 - 3y - A(x, y))dx + dy = 0$

(a) A Equação diferencial é uma EDO de 1ª ordem.

True

Your last answer was interpreted as follows: **True**

Correct answer, well done.

(b) Para $A(x, y) = x^2 - 3$:

(i) A equação diferencial não é linear e de 1ª ordem.

True

Your last answer was interpreted as follows: **True**

Incorrect answer.

(ii) Determine a solução geral da ED e introduza a constante com %c.

$$y = f(x; c) \Leftrightarrow$$

(c) Para $A(x, y) = 0$

(i) determine a solução particular da equação diferencial que satisfaz a a condição inicial $y(0) = 4$.

$$y = f(x) \Leftrightarrow$$

(ii) Sendo a figura 1 o gráfico e campo direcional da ED, qual das figuras 2 ou 3 é o gráfico da sua solução geral?

Figura = 2

Your last answer was interpreted as follows: 2

(ii) Sendo a figura 1 o gráfico e campo direcional da ED, qual das figuras 2 ou 3 é o gráfico da sua solução geral?

Figura =

Your last answer was interpreted as follows: 2

Correct answer, well done.

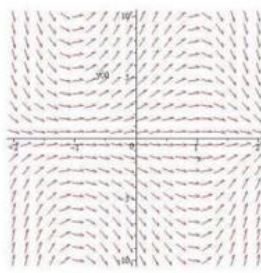


Figura 1

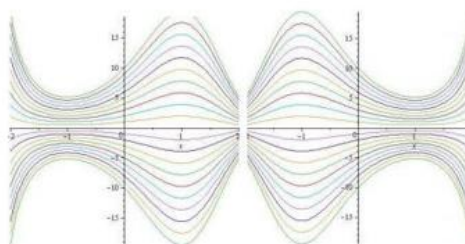


Figura 2

Figura 3

Question 4
Partially correct
Mark 1.000 out of 2.000
Flag question

Considere o sistema de funções $SF = \{\sin(6 \cdot x), \cos(6 \cdot x)\}$.

a) Calcule o Wronskiano do sistema de funções SF.

$\det(W) = -6$

Your last answer was interpreted as follows: -6

b) SF constitui um Sistema Fundamental de Soluções (SFS) de uma equação diferencial de ordem 2, linear e homogênea?

True

Your last answer was interpreted as follows: True

c) As funções de SF são soluções da equação diferencial $y'' + 6y = 0$.

True

Your last answer was interpreted as follows: True

d) Determine a solução geral da equação diferencial $y'' + 36y = 0$.

$y = c_1 * [y$

Your last answer was interpreted as follows: y

The variables found in your answer were: [y]

$+ c_2 * [6 * y^3$

Your last answer was interpreted as follows: $6 \cdot y^3$

The variables found in your answer were: [y]

com $c_1, c_2 \in \mathbb{R}$.

Question 5

Incorrect
Mark 0.000 out of 2.000
Flag question

Considere o PVI de ordem 2 definido por:

$$(P) \begin{cases} y'' - 4y = 0 \\ y(0) = 1 \\ y'(0) = 0 \end{cases}$$

a) Determine a solução particular de P.

$$y = y(t) \Leftrightarrow \frac{2}{3}y^3 + \%c2 + \%c1 \cdot y$$

Your last answer was interpreted as follows: $\frac{2}{3} \cdot y^3 + \%c2 + \%c1 \cdot y$

The variables found in your answer were: $[\%c1, \%c2, y]$

b) Transforme o problema diferencial P num PVI de ordem 1, isto é, com um sistema de duas equações diferenciais de ordem 1.

$$(Q) \begin{cases} u' = f(t, u, v) \\ v' = g(t, u, v) \\ u(0) = 1 \\ v(0) = 0 \end{cases}$$

$$f(t, u, v) = 2y^2 + \%c$$

Your last answer was interpreted as follows: $2 \cdot y^2 + \%c$

The variables found in your answer were: $[\%c, y]$

$$g(t, u, v) = y \%c$$

Your last answer was interpreted as follows: $y \cdot \%c$

The variables found in your answer were: $[\%c, y]$

Question 6

Partially correct
Mark 2.400 out of 4.000
Flag question

Considere as funções reais de duas variáveis reais definidas por:

$$f(x, y) = y^2 + x^2, \quad g(x, y) = -\frac{1}{4}(y^2 + x^2), \quad \text{se } x^2 + y^2 \leq 16, \quad h(x, y) = -2\sqrt{x^2 + y^2} + 4, \quad \text{se } 1 \leq x^2 + y^2 \leq 4$$

a) Determine as derivadas parciais seguintes:

$$\frac{\partial h}{\partial x}(x, y) = \frac{(-2 \cdot x)}{\sqrt{x^2 + y^2}}$$

Your last answer was interpreted as follows: $\frac{-2 \cdot x}{\sqrt{x^2 + y^2}}$

The variables found in your answer were: $[x, y]$

Correct answer, well done.

$$\frac{\partial g}{\partial y}(x, y) = \frac{-(y/2)}{}$$

Your last answer was interpreted as follows: $-\frac{y}{2}$

The variables found in your answer were: $[y]$

Correct answer, well done.

b) Determine a equação da reta tangente à curva C de interseção da superfície de equação $z = g(x, y)$ com o plano $x = 2$ no ponto $P(x, y) = (2, 2)$.

i) Qual é o declive da reta tangente à curva C no ponto P ? $m_t = -1$

Your last answer was interpreted as follows: -1

Correct answer, well done.

ii) A equação da reta tangente é dada por: $x = 2 \wedge z = -y - 2$

Your last answer was interpreted as follows: $-y - 2$

The variables found in your answer were: $[y]$

Incorrect answer.

c) A temperatura de uma placa de metal aquecida é dada por $T(x, y) = y^2 + x^2$.
Determine a taxa de variação de T em relação à distância no ponto no ponto $P(x, y) = (2, 2)$ na direção:

i) do eixo dos $xx =$

Your last answer was interpreted as follows: 4

Correct answer, well done.

ii) do eixo dos $yy =$

Your last answer was interpreted as follows: 4

Correct answer, well done.

iii) do vetor que faz um ângulo de 30° com a direção positiva do eixo dos $xx =$

d) Se $z = 4 - 2 \cdot \sqrt{y^2 + x^2}$, $x = \rho \cos(\theta)$, $y = \rho \sin(\theta) \Rightarrow \frac{\partial^2 z}{\partial \theta^2} + \frac{\partial z}{\partial \rho} = \left(\frac{\partial z}{\partial y}\right)^2 + \left(\frac{\partial z}{\partial x}\right)^2$

True

Your last answer was interpreted as follows: **True**

Incorrect answer.

e) O domínio da função h é um círculo fechado.

False

Your last answer was interpreted as follows: **False**

Correct answer, well done.

f) Das figuras seguintes qual delas é o gráfico da função:

i) $z = g(x, y) \rightarrow$ Figura =

Your last answer was interpreted as follows: 3

ii) Das figuras seguintes qual delas é o gráfico da função:

i) $z = g(x, y) \rightarrow$ Figura =

Your last answer was interpreted as follows: 3

Correct answer, well done.

ii) $z = h(x, y) \rightarrow$ Figura =

Your last answer was interpreted as follows: 1

Correct answer, well done.

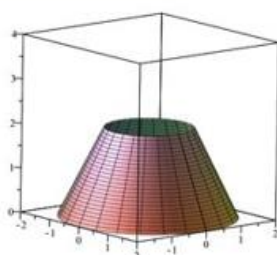


Figura 1

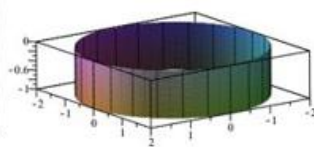


Figura 2

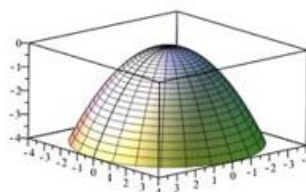
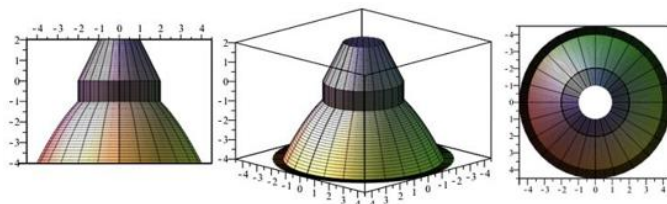


Figura 3

A figura seguinte representa um protótipo de um aparelho para afastar a D.Toupeira da horta EcoCampus!SEC.
O sólido é composto por 4 partes/superfícies, a saber:

- Tronco de um cone de altura $h = 4$ e raio $r = 2$;
- Cilindro de raio $r = 2$ e altura $h = 1$;
- Segmento de um paraboloide de altura $h = 4$ e largura máxima de raio $r = 4$;
- Anel circular de largura $l = \frac{1}{2}$.



(a) Associando os conjuntos seguintes a sistemas de coordenadas 3D, complete-os de forma a definir corretamente o sólido $S = S_1 \cup S_2 \cup S_3 \cup S_4$:

$$S_1 = \{(\rho, \theta, z) : r_1 \leq \rho \leq 2 \wedge 0 \leq \theta \leq \theta_2 \wedge z = z(\rho, \theta)\}$$

$$r_1 = \boxed{0}$$

Your last answer was interpreted as follows: 0

Incorrect answer.

$$\theta_2 = \boxed{2\pi}$$

Your last answer was interpreted as follows: $2 \cdot \pi$

Correct answer, well done.

$$z(\rho, \theta) = \boxed{-2\sqrt{x^2+y^2}+4}$$

Your last answer was interpreted as follows: $-2 \cdot \sqrt{x^2 + y^2} + 4$

The variables found in your answer were: $[x, y]$

$$S_2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \leq 4 \wedge z_1 \leq z \leq 0\}$$

$$z_1 = \boxed{-1}$$

Your last answer was interpreted as follows: -1

Correct answer, well done.

$$S_3 = \{(\rho, \theta, z) : r_1 \leq \rho \leq r_2 \wedge 0 \leq \theta \leq 2\pi \wedge z = z(\rho, \theta)\}$$

$$r_1 = \boxed{0}$$

Your last answer was interpreted as follows: 0

Incorrect answer.

$$r_2 = \boxed{4}$$

Your last answer was interpreted as follows: 4

Correct answer, well done.

$$z(\rho, \theta) = \boxed{-(x^2+y^2)}$$

Your last answer was interpreted as follows: $-(x^2 + y^2)$

The variables found in your answer were: $[x, y]$

Incorrect answer.

$$S_4 = \{(x, y, z) : r_1 \leq x^2 + y^2 \leq r_2 \wedge z = z(x, y)\}$$

$$r_1 = \boxed{-8/2}$$

Your last answer was interpreted as follows: $-\frac{8}{2}$

Incorrect answer.

$$r_2 = \boxed{-4}$$

Your last answer was interpreted as follows: -4

Incorrect answer.

$$z(x, y) = \boxed{-4}$$

Your last answer was interpreted as follows: -4

Correct answer, well done.

(b) Atendendo à forma do candeeiro em funil, determine o volume de vinho de Pinhel (Cidade do vinho 2020-21) do funil cheio (capacidade do funil) e a massa do anel inferior sabendo que a sua densidade é $\rho(x, y) = 7$

Nota: por uma questão de simplificação dos cálculos para o cálculo do volume de vinho, considere que a espessura do funil é desprezável.

$$(i) V(S) = V(S_1) + V(S_2) + V(S_3)$$

$$V(S_1) = \boxed{}$$

$$V(S_2) = \boxed{4\pi}$$

Your last answer was interpreted as follows: $4 \cdot \pi$

Correct answer, well done.

$$V(S_3) = \boxed{}$$

$$(ii) M(S_4) = \boxed{7 \cdot \text{Volume}}$$

Your last answer was interpreted as follows: $7 \cdot \text{Volume}$

This answer is invalid. Forbidden variable or constant: **Volume**.

(c) Defina S_4 em coordenadas cilíndricas completando o conjunto seguinte:

$$S_4 = \{(\rho, \theta, z) : \rho_1 \leq \rho \leq \rho_2 \wedge \theta_1 \leq \theta \leq \theta_2 \wedge z = z(\rho, \theta)\}$$

$$\rho_1 = \boxed{-8/2}$$

Your last answer was interpreted as follows: $-\frac{8}{2}$

Incorrect answer.

$$\rho_2 = \boxed{-4}$$

Your last answer was interpreted as follows: -4

$$\theta_1 = \boxed{0}$$

Your last answer was interpreted as follows: 0

Correct answer, well done.

$$\theta_2 = \boxed{2\pi}$$

Your last answer was interpreted as follows: $2 \cdot \pi$

Correct answer, well done.

$$z(\rho, \theta) = \boxed{-4}$$

Your last answer was interpreted as follows: -4

Correct answer, well done.

(d) A expressão seguinte permite determinar o volume do tronco de cone S_1 .

$$I = \int_0^2 \int_0^{2\pi} \int_0^{-2\rho+4} r \, dz d\theta d\rho - \int_0^1 \int_0^{2\pi} \int_2^{-2\rho+4} r \, dz d\theta d\rho$$

True

Your last answer was interpreted as follows: **True**

Question 8Complete
Mark 1,000 out
of 2,000Flag
question

Complete as funções seguintes implementadas em MATLAB e associe-as a métodos numéricos para a resolução de problemas diferenciais.

a)

```
1 function [t,u,v] = MN01(f,g,a,b,n,u0,v0)
2
3 h = _____;
4
5 t = _____;
6
7 u = zeros(____,____);
8
9 v = _____;
10
11 u(1) = _____;
12
13 v(1) = _____;
14
15 for i = 1:n
16
17     u(i+1) = u(i)+h*f(t(i),u(i),v(i));
18
19     v(i+1) = v(i)+h*g(____);
20
21     u(i+1) = _____;
22
23     v(i+1) = _____;
24
25 end
```

b)

```
1 function [x,y,d2ydx2]=MN02(f,a,b,h,y)
2
3 x = _____;
4
5 n = _____;
6
7 if nargin==4 y=f(____); end
8
9 d2ydx2 = zeros(____,____);
10
11 for i=2:n-1
12
13     d2ydx2(i) = (____)/h^2;
14
15 end
16
17 d2ydx2(1)= _____;
18
19 d2ydx2(n)= _____;
```